Layerview

ME 395 Final Presentation, 3-12-2025

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Problem Statement & Motivation

- Apply fringe projection technology (FPP) to monitor an FDM 3D printing process, providing an automated accuracy report and part reconstruction to rectify errors when applicable.

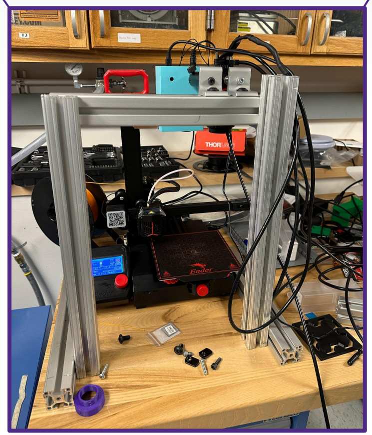
- Sustainability, Accuracy, Efficiency
  - Conserve material, money, time
  - Improve printing accuracy and rectify errors
  - Reduce need for reprint

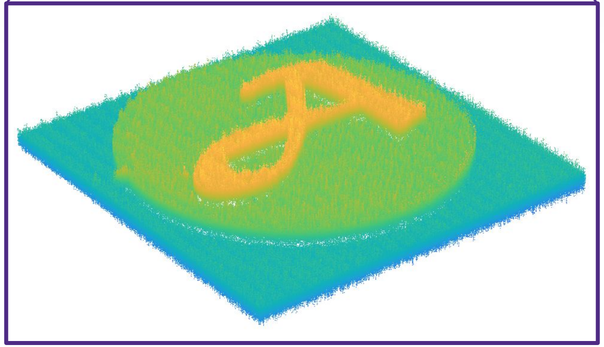
Approach

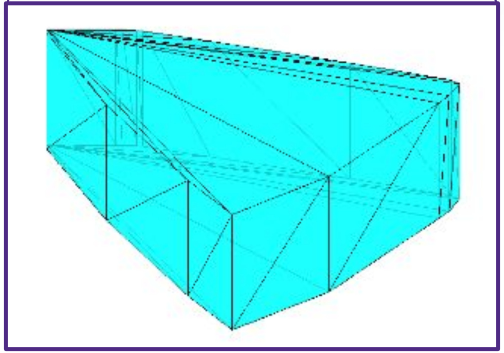
Step 1: Print and scan the part on the Ender 3 Pro using fringe projection

Step 2: Process the fringe projection data and align using ICP, and provide an error report

Step 3: Develop an STL file based on the "missing geometry" and fix the part using the printer

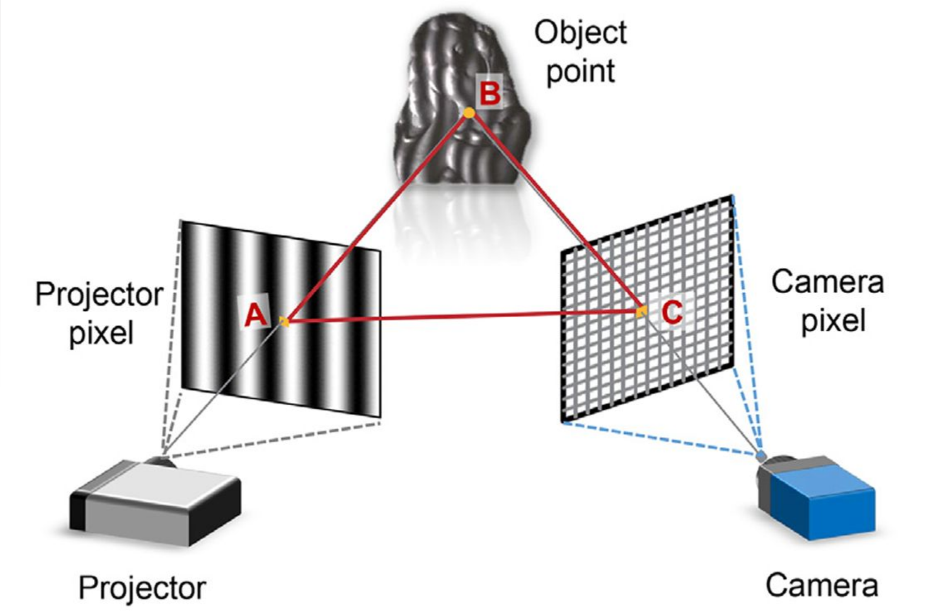






Step 1: Scanning and Printing the Part

Fringe Projection Profilometry



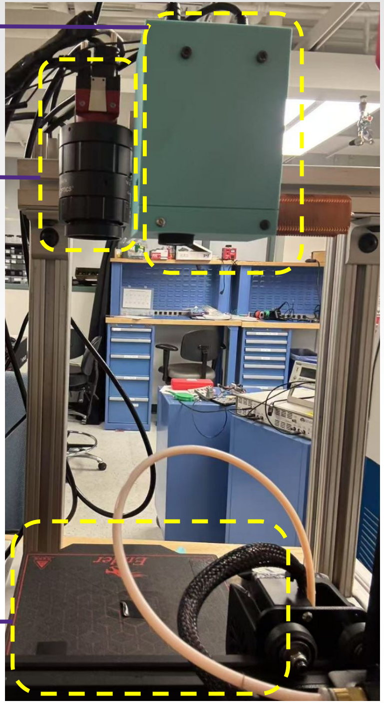
Schematic of Fringe Projection Profilometry

Our Setup

Projector

Camera

Object Platform



Feng, Shijie, et al. "Calibration of fringe projection profilometry: A comparative review." *Optics and lasers in engineering* 143 (2021): 106622.

Calculate Phase Distribution

Extract Phase

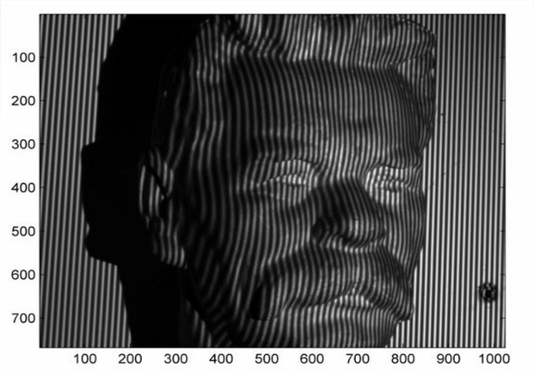
Remove Phase Jumps

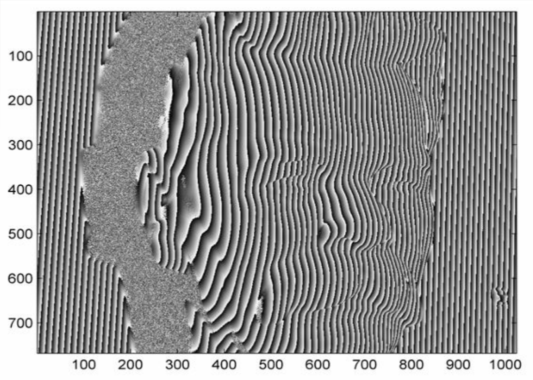
Distorted Fringe Patterns

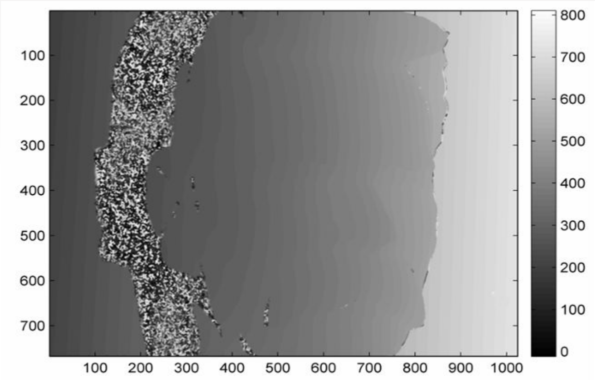
Wrapped Phase (-π to π)

Unwrapped Phase (Continuous)

Y Pixel







Phase

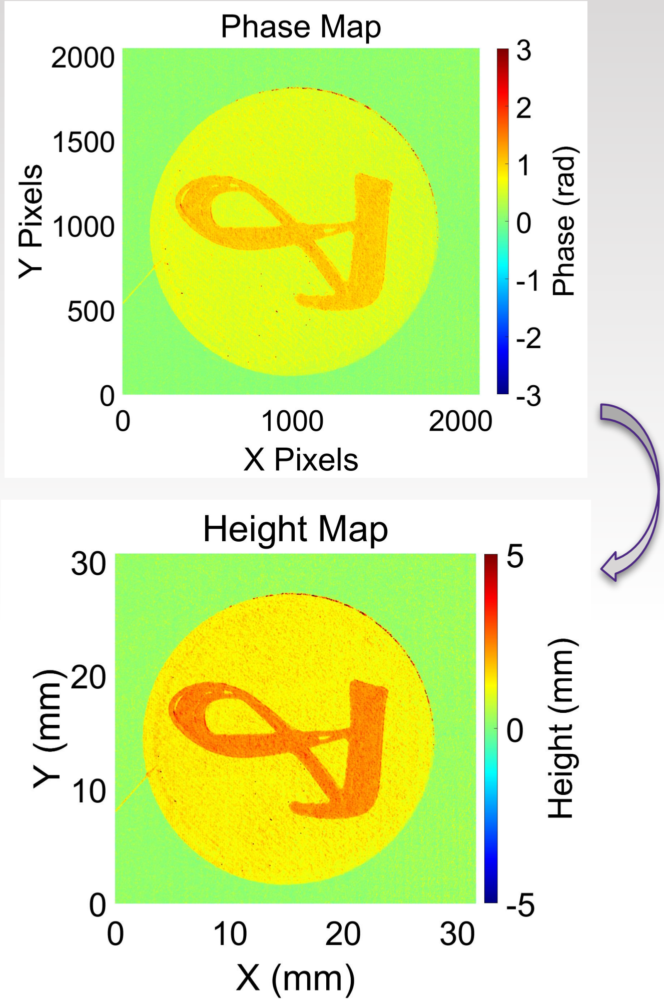
X Pixel

Li, Zhongwei, and Congjun Wang. "A prototype system for high precision 3D measurement based on grating method." *Optical Design and Testing III*. Vol. 6834. SPIE, 2007.

Calibration

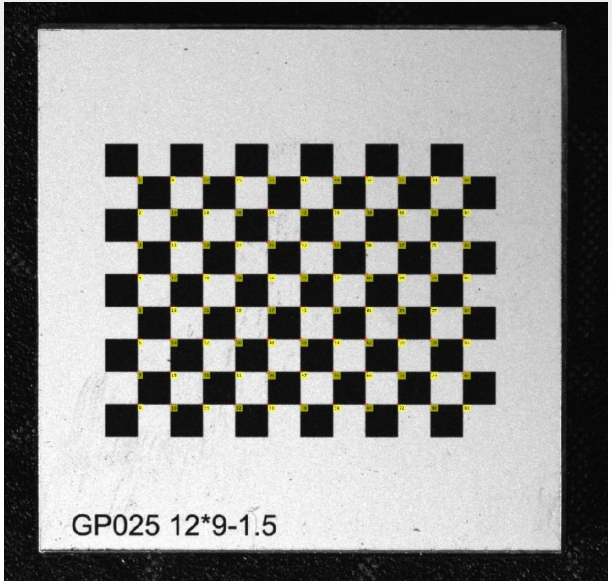
- **Phase to Height**
  - A nonlinear least squares problem
  - Solved using Levenberg-Marquardt algorithm

- **XY Pixels to Real-World Coordinates**
  - Assumes no camera distortion for simplification
  - A linear least squares problem
  - Solved using Homography Matrix

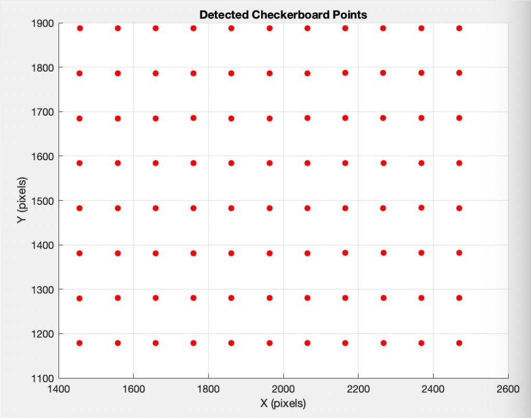


Calibration: Homography Matrix

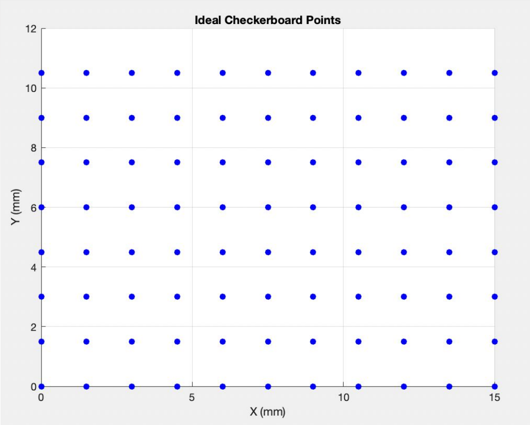
Matlab detectCheckerboardPoints function to locate corners (in pixels)



Matlab detected points (pixels)



Ideal checkerboard points (inches)

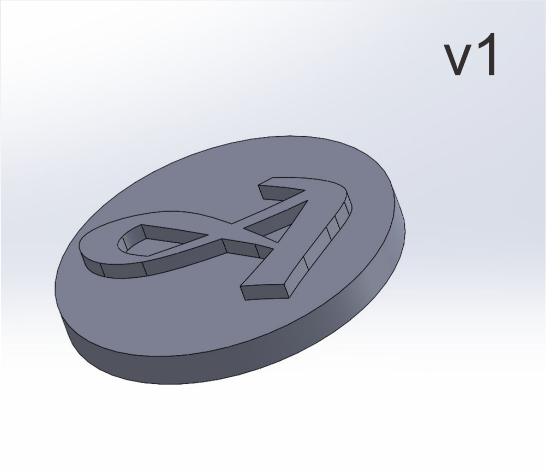


Use SVD method to calculate matrix that represents the transformation. Apply this matrix to all future point cloud scans.

Part Design

14min print time → 4min print time

First Party Creality Slicer



13+6 layers, 0.2mm layer height, 100% infill

3+3 layers, 0.2mm layer height, 100% infill

1 inch Diameter to fit FoV of Camera

G-Code Modification for Scanning

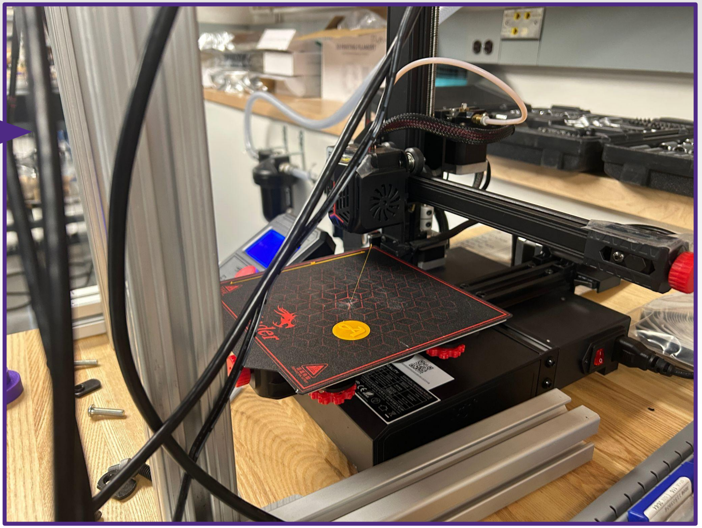
Linear move

```
G1 X0 Y165 F3000 ; Move extruder to home corner before pause
M104 S0 ;Turn-off hotend to prevent overheat
M0 ; Pause for photo
M104 S200
M109 S200
```

Changes for Base Coin Scan

Set hotend temp

Wait for hotend to reach target temp



Printer Scan Position

Extruder retracted to the left, while base plate is extended to the fullest to allow uninhibited scan

Changes for final scan

```
G1 X0 Y165 F3000 ; Move extruder to home corner before finishing
```

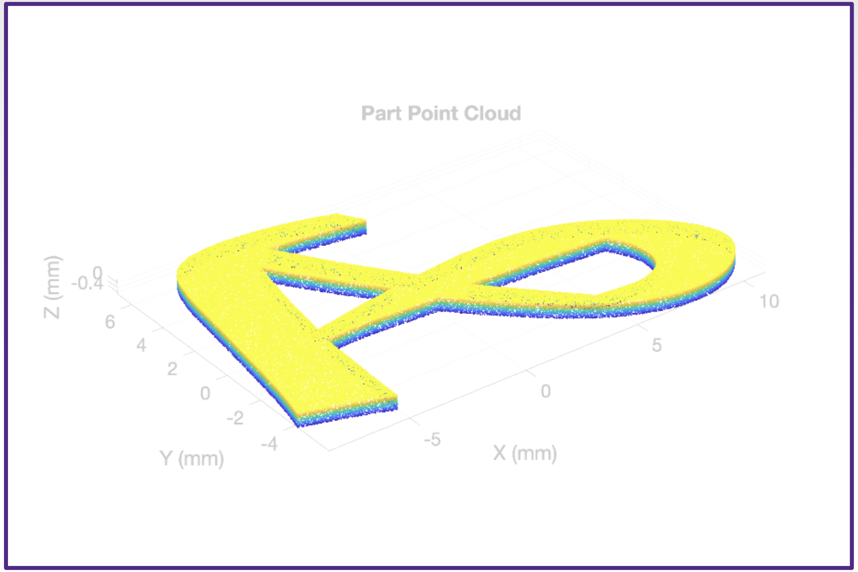
Linear move

Step 2: Processing the Scan Data

STL Data Processing

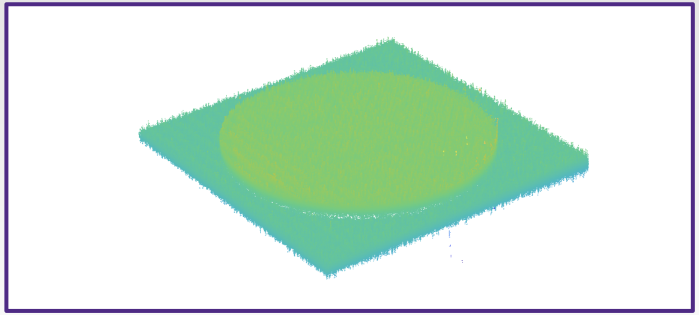
- To create the known geometry, the STL is loaded into MATLAB, with random points added along the walls of the part

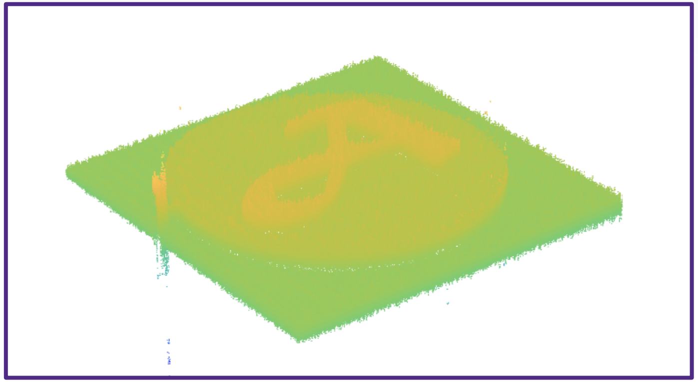
- The known geometry of the base is used to remove unneeded points



Initial Filtering and Alignment

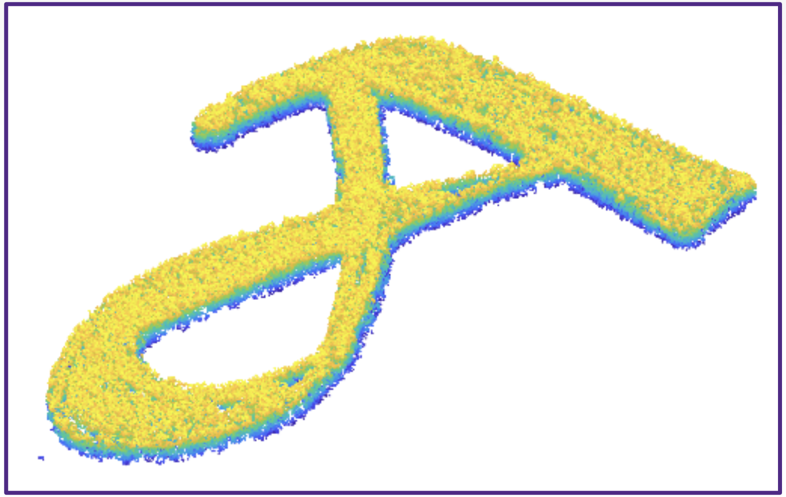
- First the scan of the base is used to align the part scan in z
- Initial filtering is applied using statistical outlier removal
- A significant amount of noise in the data remains requiring more advanced filtering techniques





Primary Filtering

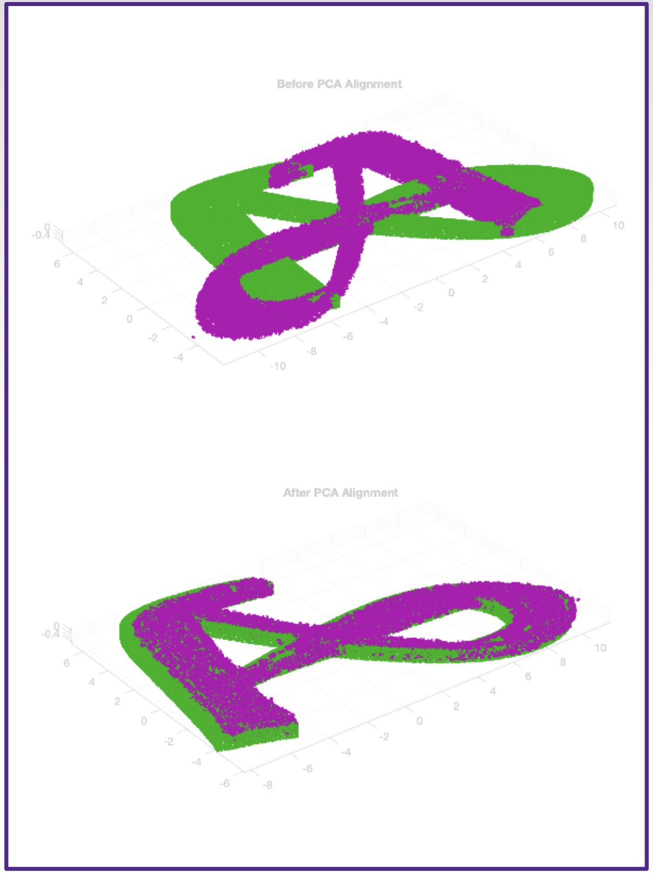
- To remove outliers, k nearest neighbors filtering is applied
- The median of the maximum (z) 25% of points in the initial scan of the base is then used to separate the base from the part
- K nearest neighbors is applied again with a larger number of neighborhood points and smaller threshold



PCA Alignment

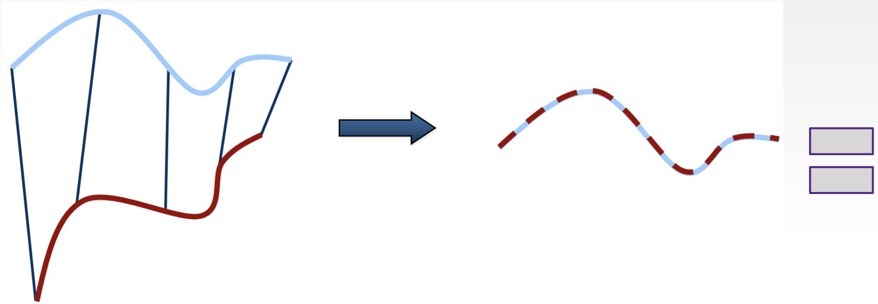
- In the final steps, downsampling based on a grid average is used to remove outliers

- PCA is used to rotate the two STL files to the same orientation
  - Each shape has ≅ eigenvectors
  - Rotate the shapes to their principal directions then check error at Θ and Θ + 180°



Iterative Closest Point Algorithm (ICP)

Given 2 point clouds, “point to plane” ICP performs translation & rotation to minimizes the difference (through root mean squared error)



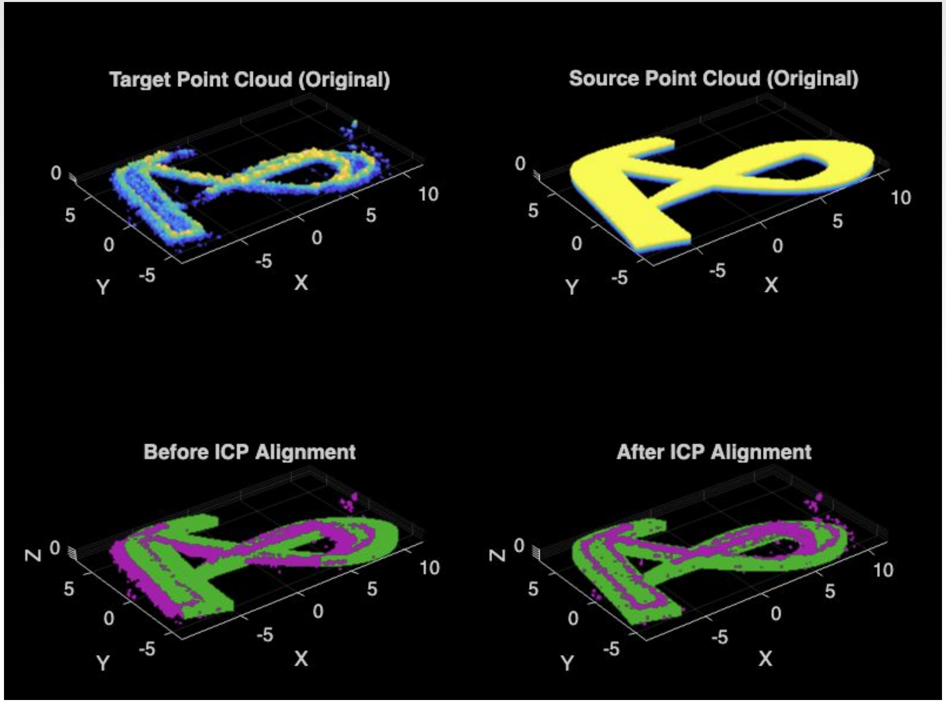
Estimated Transformation Matrix:

1.0000	-0.0020	0.0012	0
0.0020	1.0000	0.0004	0
-0.0012	-0.0004	1.0000	0
-0.0823	0.0446	-0.0079	1.0000

Final Registration RMSE: 0.050013

In MATLAB, ICP outputs the transformation matrix and root mean squared error

Error ICP



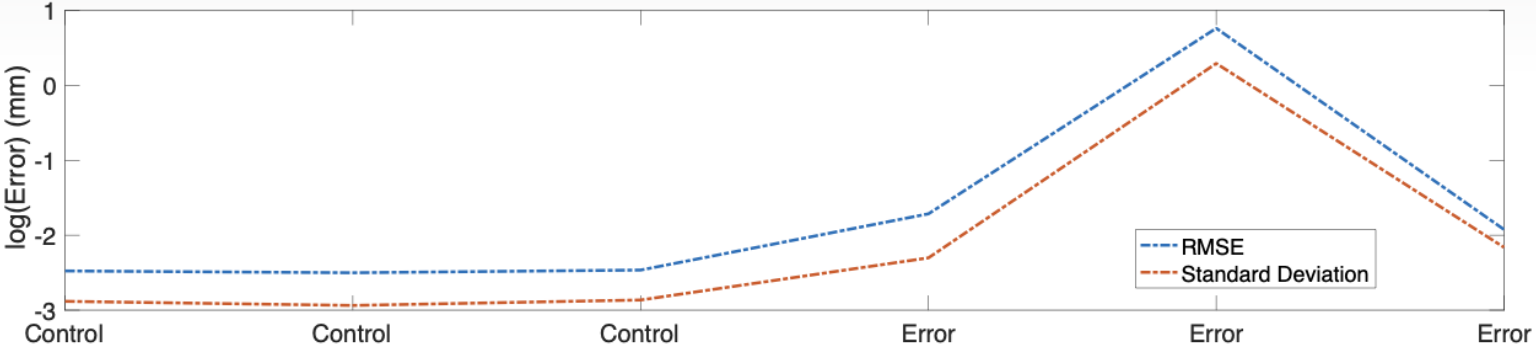
Estimated Transformation Matrix:

1.0000	-0.0000	0.0069	0
0.0000	1.0000	0.0008	0
-0.0069	-0.0008	1.0000	0
-1.2936	0.2814	-0.0270	1.0000

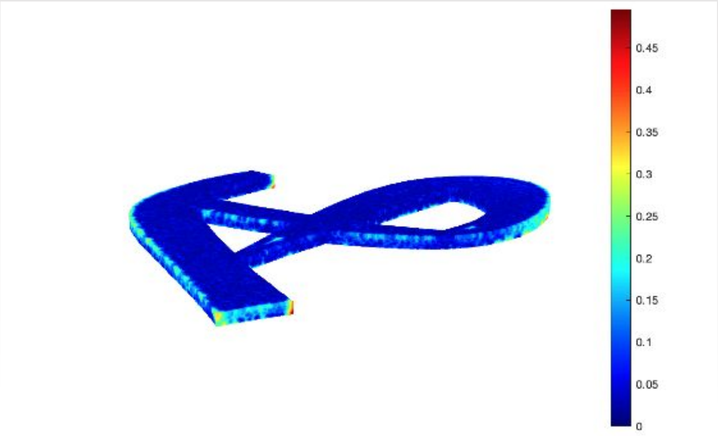
Final Registration RMSE: 0.17601

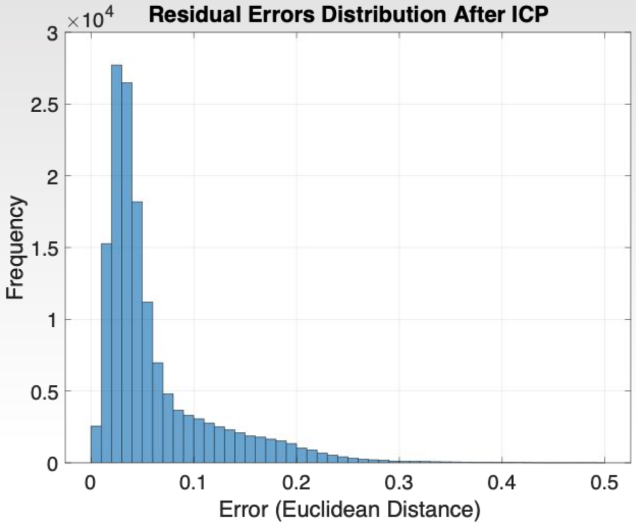
Determining Baseline Error

Part	Control	Control	Control	Defect 1	Defect 2	Defect 3
RMSE (mm)	0.084	0.082	0.085	0.180	2.143	0.146
Standard Deviation (mm)	0.056	0.053	0.057	0.100	1.340	0.115



Baseline Data Visualization



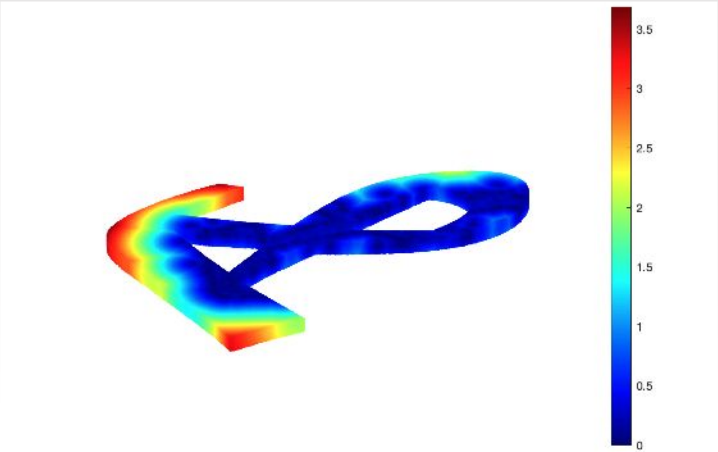


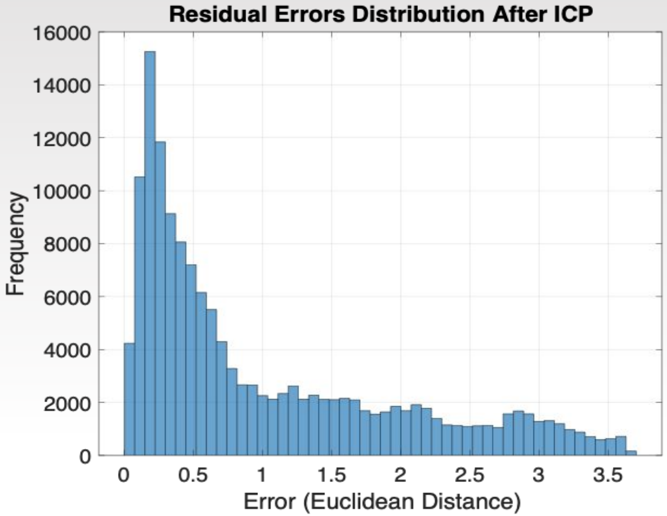
Error heat map: after ICP the aligned point cloud is compared to the reference point cloud (stl) by calculating the Euclidean distance between the corresponding points

Error histogram: provides a statistical analysis of the distribution of errors

- x-axis represents the error magnitude (in Euclidean distance)
- y-axis represents the frequency (number of points in error range)

Error Data Visualization





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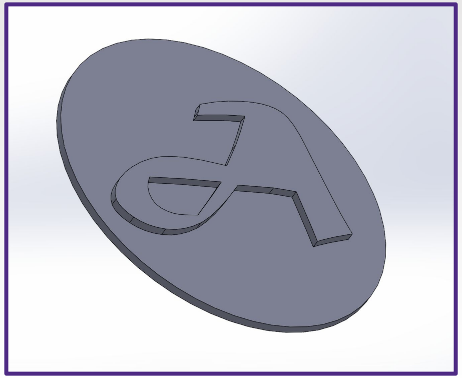
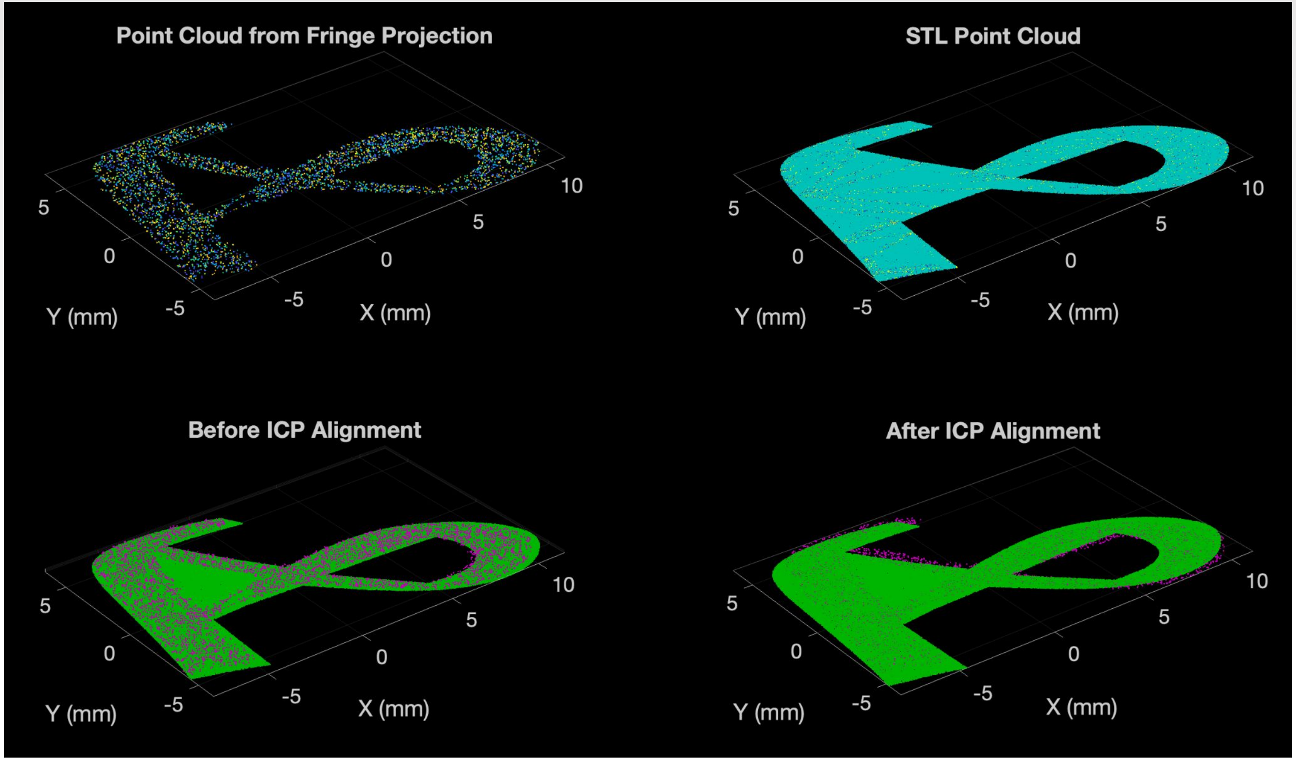
Step 3: Error Correction

Surface Alignment

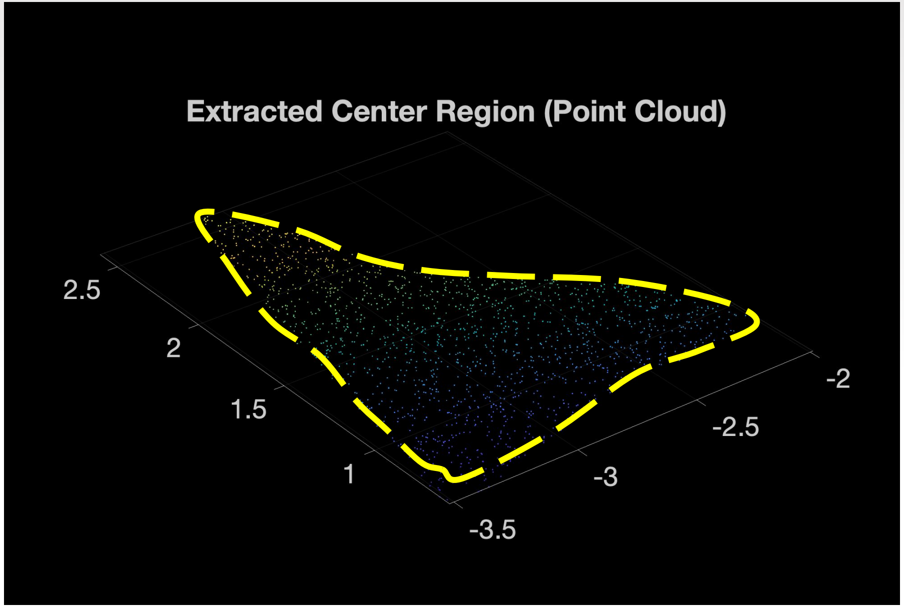




Subtracting Point Clouds

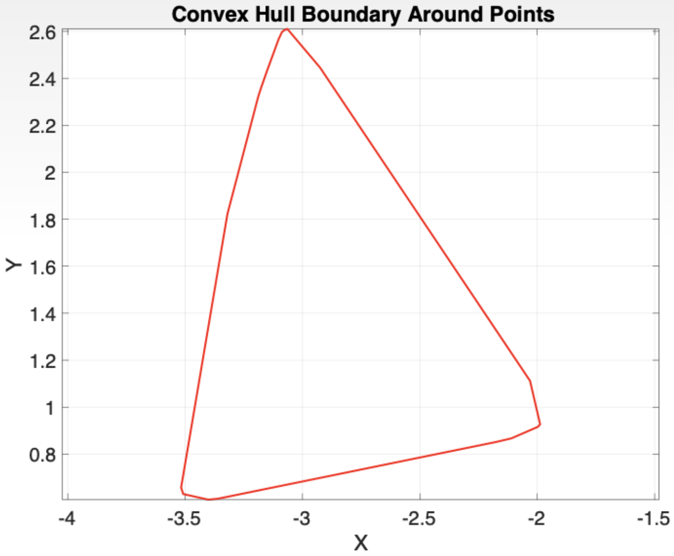
- The aligned point clouds are “subtracted” using a k-nearest neighbor search

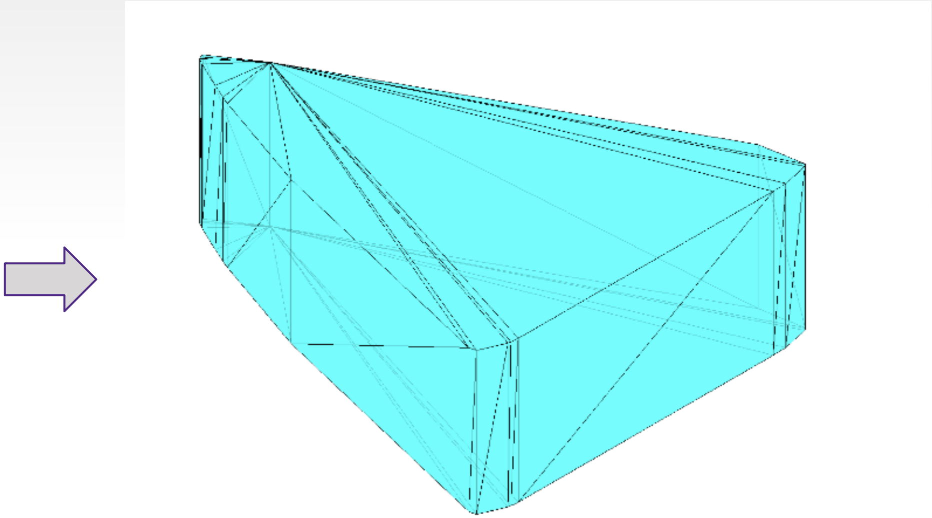
- Points in the scan that are not near other points are kept, representing the defect in 2D



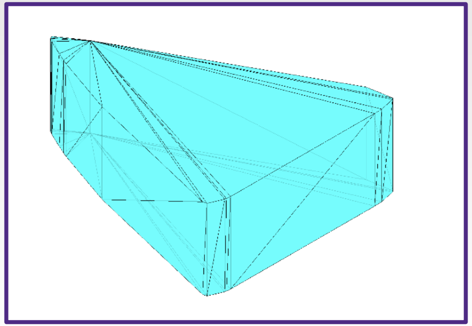
STL Construction

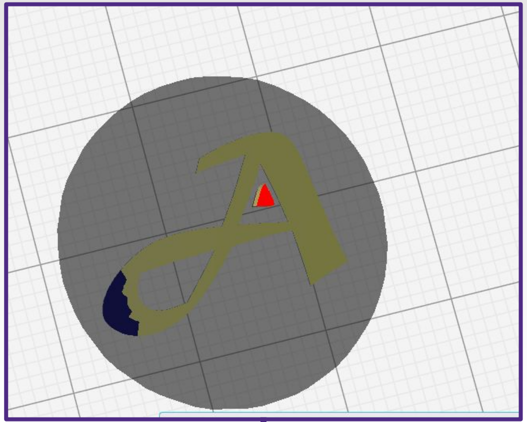
- The convex hull function is used to define the geometry for extrusion, with the resulting edges providing the base for triangulation

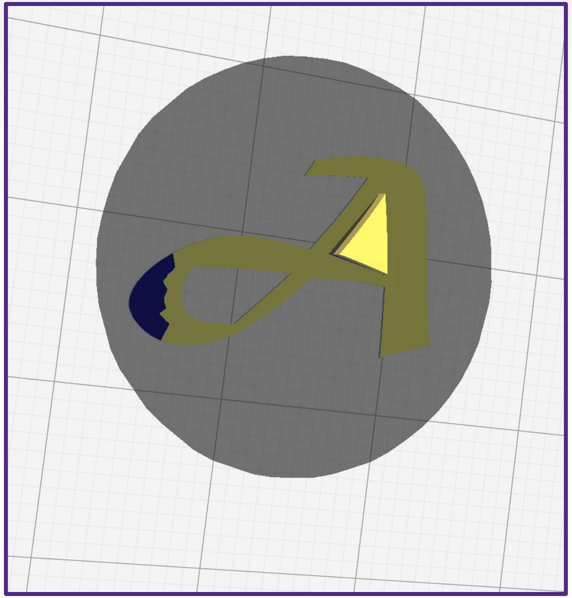




Print Repair Process







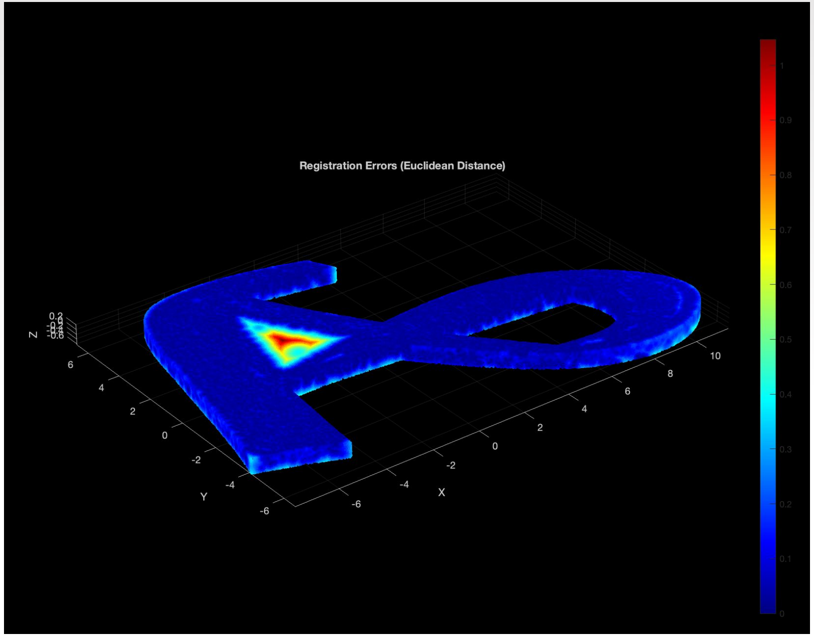
Transformation

```
G0 F6000 X120.716 Y64.173 Z0.8
```

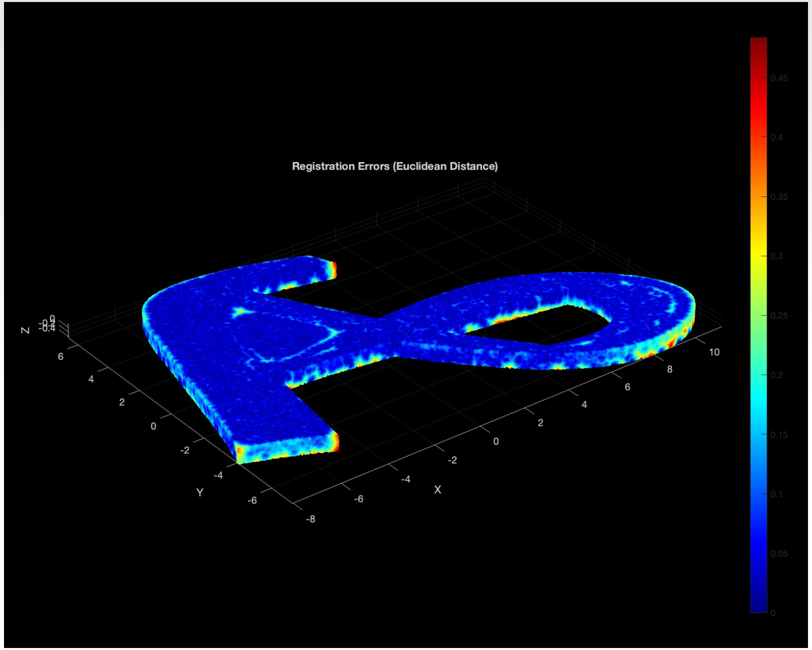
Insert z-axis translation into the G-Code

Print Repair Process

Repair Results



RMSE = 0.15 mm



RMSE = 0.10 mm

Thank you!

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References

[1] Smistad, E., Falch, T. L., Bozorgi, M., Elster, A. C., & Lindseth, F. (2014). Medical image segmentation on GPUs – A comprehensive review. *Medical Image Analysis*, *20*(1), 1–18. https://doi.org/10.1016/j.media.2014.10.012

[2] Rusu, R. B., Z. C. Marton, N. Blodow, M. Dolha, and M. Beetz. "Towards 3D Point Cloud Based Object Maps for Household Environments". Robotics and Autonomous Systems Journal. 2008.

[3] Besl, P.J., and Neil D. McKay. "A Method for Registration of 3-D Shapes." IEEE Transactions on Pattern Analysis and Machine Intelligence 14, no. 2 (February 1992): 239–256. https://doi.org/10.1109/34.121791.

[4] W. Keller, J. Girard, M. W. Goldberg, and S. Zhang, "Precise calibration for error detection and correction in material extrusion additive manufacturing using digital fringe projection," Measurement Science and Technology, vol. 36, no. 2, p. 025203, doi: https://doi.org/10.1088/1361-6501/ada847.